

Documento Entidade-Relacionamento

Plataforma Mobile de Doacoes - MongoDB/Mongoose

Resumo

Este documento foi gerado a partir dos schemas Mongoose em `src/domains/*/schemas`. O diagrama completo em Mermaid fica em `docs/diagrama-entidade-relacionamento.mmd`.

Como o projeto usa MongoDB, as relacoes sao mantidas por campos `ObjectId`. Relacoes polimorficas dependem de campos como `targetType` e `authorType`.

The diagram illustrates a database schema for a charity management system. The central entity is **USERS**, which is connected to numerous other tables. The tables and their fields are as follows:

- USERS**: Objectid, _id_PK, Date, Name, Surname, Email, Phone, Address, City, PostalCode, Sex, Gender, ProfilePicture, Bio, LastLogin, PasswordHash, Role, UserRole, IsActive, CreatedAt, UpdatedAt.
- CATEGORIES**: Objectid, _id_PK, Name, Type, Description, Status, CreatedAt, UpdatedAt.
- INSTITUTIONS**: Objectid, _id_PK, Name, Logo, Address, City, PostalCode, Website, Email, Phone, Description, Status, CreatedAt, UpdatedAt.
- CAMPAIGNS**: Objectid, _id_PK, InstitutionId_FK, Name, Description, StartDate, EndDate, Status, Progress, Visibility, StartAt, EndAt, Objectid, _id_PK, Name, Description, Status, CreatedAt, UpdatedAt.
- DONATIONS**: Objectid, _id_PK, DonorId_FK, RecipientId_FK, CampaignId_FK, Amount, Currency, Status, Visibility, StartAt, EndAt, Objectid, _id_PK, Name, Description, Status, CreatedAt, UpdatedAt.
- PAGERS**: Objectid, _id_PK, Name, Description, Status, Visibility, StartAt, EndAt, Objectid, _id_PK, Name, Description, Status, CreatedAt, UpdatedAt.
- CONVERSATIONS**: Objectid, _id_PK, Topic, Participants, InstitutionId_FK, Objectid, _id_PK, Name, Description, Status, CreatedAt, UpdatedAt.
- FOLLOWS**: Objectid, _id_PK, FollowerId_FK, FollowingId_FK, Objectid, _id_PK, Name, Description, Status, CreatedAt, UpdatedAt.
- MESSAGES**: Objectid, _id_PK, SenderId_FK, RecipientId_FK, Content, MessageType, Attachments, ReadAt, Status, CreatedAt, UpdatedAt.
- POST_REACTIONS**: Objectid, _id_PK, PostId_FK, ReactionId_FK, Content, Status, CreatedAt, UpdatedAt.
- POST_COMMENTS**: Objectid, _id_PK, PostId_FK, CommentId_FK, Content, Status, CreatedAt, UpdatedAt.
- TRACKING_EVENTS**: Objectid, _id_PK, EventId_FK, Location, Description, Status, CreatedAt, UpdatedAt.
- DONATION_STATUS_HISTORY**: Objectid, _id_PK, DonationId_FK, Status, CreatedAt, UpdatedAt.
- DELIVERY_PROOFS**: Objectid, _id_PK, DonationId_FK, Description, Status, CreatedAt, UpdatedAt.
- PWA_RECORDS**: Objectid, _id_PK, DonationId_FK, RecordId_FK, Name, Description, Status, CreatedAt, UpdatedAt.
- PAYMENTS**: Objectid, _id_PK, DonationId_FK, PaymentId_FK, Amount, Currency, Status, CreatedAt, UpdatedAt.

The relationships between the tables are as follows:

- USERS** has **CATEGORIES**.
- USERS** has **INSTITUTIONS**.
- USERS** has **CAMPAIGNS**.
- USERS** has **DONATIONS**.
- USERS** has **PAGERS**.
- USERS** has **CONVERSATIONS**.
- USERS** has **FOLLOWS**.
- USERS** has **MESSAGES**.
- USERS** has **POST_REACTIONS**.
- USERS** has **POST_COMMENTS**.
- USERS** has **TRACKING_EVENTS**.
- USERS** has **DONATION_STATUS_HISTORY**.
- USERS** has **DELIVERY_PROOFS**.
- USERS** has **PWA_RECORDS**.
- USERS** has **PAYMENTS**.
- INSTITUTIONS** has **CAMPAIGNS**.
- CAMPAIGNS** has **DONATIONS**.
- DONATIONS** has **PAGERS**.
- PAGERS** has **CONVERSATIONS**.
- CONVERSATIONS** has **FOLLOWS**.
- FOLLOWS** has **MESSAGES**.
- MESSAGES** has **POST_REACTIONS**.
- POST_REACTIONS** has **POST_COMMENTS**.
- POST_COMMENTS** has **TRACKING_EVENTS**.
- TRACKING_EVENTS** has **DONATION_STATUS_HISTORY**.
- DONATION_STATUS_HISTORY** has **DELIVERY_PROOFS**.
- DELIVERY_PROOFS** has **PWA_RECORDS**.
- PWA_RECORDS** has **PAYMENTS**.

Collections principais

Collection	Entidade	Campos relacionais principais
users	Usuario	Referenciado por doacoes, campanhas, mensagens, notificacoes, equipe, auditoria e relatorios.
institutions	Instituicao	categoryIds, verification.verifiedByUserId
categories	Categoria	Referenciada por institutions.categoryIds.
institution_staff_memberships	Equipe da instituicao	institutionId, userId, invitedByUserId
campaigns	Campanha	institutionId, createdByUserId
donations	Doacao	donorUserId, institutionId, campaignId
payments	Pagamento	donationId, donorUserId, institutionId
delivery_proofs	Comprovante de entrega	donationId, confirmedByUserId
tax_receipts	Recibo fiscal	donationId, donorUserId, institutionId
donation_status_history	Historico de status	donationId, changedByUserId
tracking_events	Evento de rastreio	donationId, actorUserId
posts	Postagem	authorId, campaignId, institutionId
post_comments	Comentario	postId, userId, parentCommentId
post_reactions	Reacao	postId, userId
conversations	Conversa	participantIds, institutionId, campaignId
messages	Mensagem	conversationId, senderUserId, readBy.userId
notifications	Notificacao	userId
follows	Seguimento	followerUserId, targetId polimorfo.
reports	Denuncia	reporterUserId, reviewedByUserId, targetId polimorfo.
audit_logs	Auditoria	actorUserId, targetId polimorfo.

Relacionamentos

Origem	Destino	Campo	Cardinalidade
users	institution_staff_memberships	userId, invitedByUserId	1:N
institutions	institution_staff_memberships	institutionId	1:N
categories	institutions	categoryIds	1:N
institutions	campaigns	institutionId	1:N

Origem	Destino	Campo	Cardinalidade
users	campaigns	createdByUserId	1:N
users	donations	donorUserId	1:N
institutions	donations	institutionId	1:N
campaigns	donations	campaignId	1:N opcional
donations	payments, delivery_proofs, tax_receipts, status_history, tracking_events	donationId	1:N
users	posts, comments, reactions, messages, notifications	authorId/userId/senderUserId	1:N
conversations	messages	conversationId	1:N
users	conversations	participantIds	N:N
users	follows	followerUserId	1:N
institutions/campaigns	follows	targetId + targetType	1:N polimorfico
users	reports/audit_logs	reporterUserId/reviewedByUserId/actorUserId	1:N

Codigo Mermaid

```

erDiagram
    USERS ||--o{ INSTITUTION_STAFF_MEMBERSHIPS : userId
    INSTITUTIONS ||--o{ INSTITUTION_STAFF_MEMBERSHIPS : institutionId
    CATEGORIES ||--o{ INSTITUTIONS : categoryIds
    INSTITUTIONS ||--o{ CAMPAIGNS : institutionId
    USERS ||--o{ CAMPAIGNS : createdByUserId
    USERS ||--o{ DONATIONS : donorUserId
    INSTITUTIONS ||--o{ DONATIONS : institutionId
    CAMPAIGNS ||--o{ DONATIONS : campaignId
    DONATIONS ||--o{ PAYMENTS : donationId
    DONATIONS ||--o{ DELIVERY_PROOFS : donationId
    DONATIONS ||--o{ TAX_RECEIPTS : donationId
    DONATIONS ||--o{ DONATION_STATUS_HISTORY : donationId
    DONATIONS ||--o{ TRACKING_EVENTS : donationId
    POSTS ||--o{ POST_COMMENTS : postId
    POSTS ||--o{ POST_REACTIONS : postId
    CONVERSATIONS ||--o{ MESSAGES : conversationId
    USERS ||--o{ NOTIFICATIONS : userId
    USERS ||--o{ FOLLOWS : followerUserId
    INSTITUTIONS ||--o{ FOLLOWS : targetId
    CAMPAIGNS ||--o{ FOLLOWS : targetId

```